Product Requirements Document

AI-Enabled Gift Recommendation System MVP

Context

Our fast-fashion retail business has experienced explosive 10x growth over the past three years by capturing the U.S. teenage self-purchase market. However, we're now facing market saturation in our core demographic—the typical teenager who buys for themselves. To sustain momentum and unlock the next wave of growth, we need to shift our strategy toward leveraging the social fabric that already exists within our app. Our customers are highly engaged (NPS of 75, exceeding Amazon's), and data shows they already discuss style preferences with 2-3 close friends or family members. By building an AI recommendation engine focused on gift-giving, we can monetize this existing social behavior, increase average transaction value, and extend our peak selling season beyond traditional retail calendars.

MVP Goal

Build an AI-powered gift recommendation system that suggests relevant products for existing teenage customers to purchase as gifts for their friends and family, surfacing recommendations at key moments in the app and via email to drive incremental gifting purchases within the first holiday season.

Target Customer Segment

- Existing teenage users (ages 13-19) with demonstrated purchase history and high app engagement
- Self-purchasing behavior established: At least one prior purchase in the past 12 months to ensure brand familiarity and payment method on file
- Social network within app: Connected to 2+ friends or family members in the app; able to see or know their style preferences
- Higher-frequency purchasers: 35%+ of target customers have made 2+ purchases in the past 12 months, indicating both intent and affordability
- Low commerce friction: Existing saved addresses and credit card information reduce checkout abandonment

Timeframe

Deliver MVP recommendation engine to 100% of users by November 1st, with full holiday season (November-December) to drive adoption and measure gifting behavior change.

First 60-Day Success Metrics

Gifting Frequency

Key Performance Indicator	Target	Rationale
New gifters acquired (customers making their first gift purchase)	150,000	Represents 30% of active user base attempting gifting for the first time; strong indicator of feature relevance
Gift transactions as % of total transactions	8-12%	Establishes baseline; gifting should grow from near-zero to meaningful portion of transaction mix
Repeat gifting rate (customers who gift 2+ times in 60 days)	35%	Signals sustained behavior; if repeat rate approaches repeat purchase rate (35%), gifting becomes habitual

Gifting Value

Key Performance Indicator	Target	Rationale
Average order value for gift transactions vs. self-purchase	$5-8 higher	Gift purchases should outperform self-purchases; addressable to different person; willing to spend more
Gift transaction conversion rate	2.5-3.5%	Slightly higher than baseline self-purchase conversion (2%), indicating targeted recommendations drive action
Total gifting GMV (gross merchandise value)	$5M+	Conservative estimate based on 150K new gifters × $35 avg order value across 60 days

Overall Marketplace Value

Key Performance Indicator	Target	Rationale
Overall app transaction volume increase	+15-20%	Gifting should add incremental volume without cannibalizing self-purchase behavior
Customer lifetime value (LTV) uplift for users exposed to gifting feature	+$25-50	Gifting expands use cases and increases engagement; should measurably increase customer wallet share
Repeat purchase frequency (self + gift)	42% (vs. baseline 35%)	Combined behavior (buying for self + others) should exceed self-purchase-only baseline

AI Models and Data

Recommendations: Hybrid Collaborative & Content-Based Filtering System

Model Selection & Rationale:

We will build a Hybrid Recommendation Engine that combines:
• User-Based Collaborative Filtering (identifying friends with similar taste profiles to recommend products they've liked)
• Content-Based Filtering (matching past purchase behavior to product attributes and recommending similar items)

This hybrid approach is essential for the gifting use case because it solves two distinct problems: (1) who to buy for (via social graph and past gifting history) and (2) what they'll like (via their own purchase patterns and browsing signals). Neither approach alone captures the full picture; collaborative filtering tells us who has similar taste, but content-based filtering ensures recommendations align with product attributes the recipient has previously engaged with.

Recommendation Data Sources (Explicitly Classified):

Data Source	Classification	Explanation	Why It Enhances Output
Purchase history for self (past 12 months)	Exogenous	Historical purchases are fixed facts that occurred in the past; they don't depend on the recommendation system's previous outputs. They represent ground truth about what the customer has already done.	Enables content-based filtering: If User A purchased pastel hoodies and User B also purchased similar hoodies, we can infer they have overlapping taste. This is the foundation for saying "User A might like what User B purchased."
Past gift purchases for specific recipients	Autoregressive	Past gift purchases depend on the recipient—if we recommended a gift to User A for their friend in the past, and they purchased it, that outcome influences future recommendations for that friend. The model learns "this recipient liked this product when gifted."	Enables personalized occasion-based recommendations: If User A previously gifted a certain category to Friend X on Friend X's birthday, the model learns this is a strong signal. Future birthday recommendations for Friend X should lean heavily on this pattern.
Social connections and interests (friend list,	Exogenous	A user's friend list and language setting are	Enables recipient targeting: The model

social activity, language/locale)		external facts—they're not outputs of the recommendation system. They're static demographic/social data provided by the user.	uses the friend list to identify who to buy for, and uses language/locale to infer cultural preferences (e.g., certain holidays matter more in certain regions). This prevents recommending gifts for people outside the user's social graph.
Browsing and engagement signals (time on product page, add-to-cart, wishlist activity)	Autoregressive	Browsing behavior is dynamic and responsive to the app's product recommendations and content. If the system recommends a product and the user spends time viewing it, that engagement signal feeds back into the model to improve future recommendations. It's a feedback loop.	Enhances signal beyond purchase: Not every interested customer buys immediately. Add-to-cart and wishlist activity reveal intent even before purchase. For recipients (friends) whose wishlist the user can see, this is a direct, high-confidence signal of what to gift.
Demographic and contextual data (user age, recipient birthdate, location zip code)	Exogenous	Age and birthdate are static user attributes; location is provided by billing address. These don't change based on recommendation output.	Enables occasion-triggered and location-aware recommendations: If the system knows Friend X's birthday is in 2 weeks, it can trigger birthday-specific recommendations. If the user and recipient are in different climates, seasonal product recommendations can be regionalized.
Trending products by occasion and geography	Exogenous	Aggregate trending data (e.g., "hoodies are trending as gifts in the Northeast in December") is external—it reflects market behavior, not the	Adds market signal to personalization: While personalized recommendations are strong, they risk over-narrowing (only recommending what

		system's past recommendations. It's updated from sales data across all users, not individual user actions.	the user has seen). Trending data ensures popular gifts surface even if they're new to the recipient. This balances personalization with discovery.

Why This Classification Matters:

Exogenous data (purchase history, friend list, demographics, trends) provides stable input signals that don't change based on the model's output. These are safe to use immediately without causing feedback loops.

Autoregressive data (past gift purchases, engagement signals) represent learned patterns from prior gifting behavior. These improve over time as more gifting transactions occur, making the model increasingly accurate in subsequent iterations.

By explicitly classifying each dataset, we ensure the model can distinguish between "facts about the user" (exogenous) and "learned patterns from gifting behavior" (autoregressive), which is critical for avoiding recommendation staleness and ensuring the system improves as it gathers more gifting data.

Forecasts: Time-Series Autoregressive Model for Demand Forecasting

We will build an autoregressive forecasting model to predict gifting demand by occasion, product category, and time horizon. This enables inventory planning, marketing spend allocation, and feature prioritization. The model will track velocity of gift purchases for recurring occasions to forecast future demand.

Data sources:

- Historical gifting transactions (autoregressive): Past 12-24 months of gift purchases to establish seasonal patterns
- Occasion data (exogenous): Calendar events, holidays, school breaks, and major cultural moments
- User cohort data (exogenous): Age distribution of users and their friends to predict occasion relevance
- Competitor gifting signals (exogenous): Public data from competitor apps and social media trends
- Economic indicators (exogenous): Consumer confidence, discretionary spending patterns, and school calendar

User Stories

Generate Recommendations in the App

User Story A: Gift Recommendation Feed on Gifting Tab

In the app, when a teenager opens the "Gifts" tab, they see personalized gift recommendations ranked by likelihood of recipient interest, enabling them to discover gifting opportunities in moments of high engagement.

- The system displays 8-12 gift recommendations, each paired with a suggested recipient from the user's social graph
- Recommendations are ranked by a hybrid score: 70% recipient past purchase similarity + 20% user past gifting patterns + 10% trending in recipient's location
- Each recommendation card shows: product image, price, recipient name, reason for pick, and direct 'Gift to [Friend]' button
- Tapping the 'Gift to' button pre-fills checkout with recipient address and prompts shipping/occasion selection
- The feed refreshes every 24 hours or when new products arrive, keeping recommendations fresh

User Story B: Occasion-Triggered Recommendations via In-App Notification

When a friend's birthday or a major holiday approaches within 2 weeks, the app sends an in-app notification with 2-3 specific gift recommendations for that person, removing the friction of remembering and deciding.

- The system tracks upcoming occasions from the user's social graph: friend birthdays, holidays, and recurring occasions
- 14 days before a major occasion, a notification is triggered with 1-2 thumbnail recommendations
- Tapping the notification takes the user directly to a curated mini-feed of 6-8 recommendations specific to that occasion
- Occasion type (birthday vs. holiday) is factored into recommendations: birthday gifts are more personalized; holiday gifts trend toward popular items
- Only 1-2 notifications per month per user to avoid notification fatigue

Forecast and Display Results for Analysis

User Story C: Executive Dashboard: Gifting Performance & Forecasting

In a web-based analytics dashboard, business stakeholders can track real-time gifting performance, forecast demand by occasion, and identify emerging opportunities to optimize marketing spend and inventory allocation.

- Real-time metrics tile: Total gift transactions (MTD), new gifters acquired, gift GMV, gift attach rate, average gift order value
- Demand forecast chart: 90-day forward forecast of expected gift transactions by occasion with confidence intervals
- Category performance heatmap: Which product categories are trending as gifts; identifies hot gift categories

- Recipient segment breakdown: Visualizes which recipient demographics (friend, parent, sibling) receive gifts most often
- Conversion funnel by occasion: Tracks recommendation view → add to cart → purchase, segmented by occasion

User Story D: Weekly Stakeholder Email: Gifting Performance & Forecast Update

Every Monday morning, key stakeholders receive a curated email summarizing the prior week's gifting performance and the 4-week gifting forecast, enabling faster decision-making on inventory and promotion strategy.

- Performance summary (past 7 days): Total gift transactions, GMV, new gifters, repeat gifters, and KPIs vs. target
- Forecast highlight (next 4 weeks): Predicted gifting volume and GMV by week; identifies peak weeks
- Occasion breakdown: Which occasions are driving gifting; helps marketing team prioritize messaging
- Top 5 gift categories: Which products are converting best as gifts; informs buying team on inventory
- Forecast confidence note: Simple explanation of confidence levels based on data availability

Populate Backlog

User Story E: Gift vs. Personal Purchase Intent Capture

When a customer completes a purchase, they are asked explicitly whether the item is for themselves or as a gift, and if a gift, for whom and on what occasion. This data is stored to improve future recommendations and forecast accuracy.

- At checkout confirmation, a modal appears: 'Is this for you or a gift?'
- If 'Gift': follow-up questions appear on recipient and occasion with dropdown or free text options
- If 'Self': user skips modal and sees standard order confirmation
- Data is stored in the purchase record with explicit metadata tags including purchase_type, recipient_id, occasion
- Users can edit this information for 24 hours after purchase if corrections are needed

User Story F: Social Gifting Attribution & Recipient Profile Building

The app tracks who receives gifts from whom, building a "gifting profile" for each user that informs future recommendations and enables analysis of the social gifting network.

- When a gift is delivered, the app sends a notification to the recipient with sender name and product details
- Recipient can view the gift in a 'Gifts Received' section; optional: they can 'add to favorites' or leave a reaction
- The system stores: recipient_id, sender_id, product_id, occasion, date_received, recipient_reaction
- This creates a feedback loop: if Sarah frequently receives hoodies, the system infers preference and recommends similar items
- Over time, this builds a richer gifting profile than self-purchase data alone

Success Criteria & Next Steps

MVP success is measured by: achieving the First 60-Day Success Metrics outlined above, specifically validating that the gifting feature can acquire 150K new gifters and drive 8-12% of transactions to be gift-based within the holiday season.

Immediate next steps post-launch:

- Measure product-market fit for gifting by tracking retention of new gifters into Q1
- Analyze recommendation click-through rate and conversion rate by recipient segment
- Validate forecast accuracy by comparing predictions to actual results; refine model with real data
- Explore expansion to non-seasonal gifting occasions once holiday gifting is mature

Success unlocks future roadmap:

- Gifting recommendations for younger siblings
- Physical gift cards and e-gift options
- Group gifting (multiple friends pooling for larger gift)
- AI-powered occasion discovery